A Möbius-binomial C/F bridge and a claim-disciplined theorem-transfer atlas for Apéry-like sources

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Abstract

We isolate a direct Möbius-binomial bridge between Zagier's C and F Apéry-like sequences:

$$F(x) = (1 - 9x)^{-1} C\left(\frac{-x}{1 - 9x}\right).$$

Equivalently, the bridge is an involutive binomial coefficient transform between the two catalog-normalized coefficient sequences and preserves their ordinary Hankel determinants; it also transports the differential equation of C to that of F and is cross-checked by the known Franel transform route. We then use this bridge as a primitive for a theorem-transfer atlas. The atlas distinguishes closed internal results, theorem imports by reference, conditional transfers, finite exact replays, source-attached conjectures, proof obligations, blocked rows, bounded negatives, watch metadata, and closed metadata. In particular, we do not derive Sun's Conjecture 3.5 from the C/F bridge, do not promote finite replays or coefficient matches to theorems, and do not turn Dwork or Rowland–Yassawi metadata into unconditional theorem claims. Sun's raw-C conjecture is addressed in a separate Franel–Domb bridge manuscript [Bab26]. The result here is a claim-disciplined package for using a proved C/F bridge while keeping adjacent Apéry-like source layers separate.

1 Introduction

Apéry-like sequences sit at the intersection of several structures: recurrences, modular parametrizations, binomial transforms, constant-term and Laurent-polynomial models, supercongruences, and automatic congruences. These structures often appear in clusters, but they do not transfer automatically from one sequence to another. A coefficient match, a shared recurrence shape, or a common modular level may suggest a relation; none of these is by itself a theorem-transfer mechanism.

This point of view is close in spirit to the tabular literature around Calabi–Yau and Apéry-like differential equations, where large lists of operators, transformations, and periods are useful only when their normalizations and equivalences are explicit [AvEvSZ05, AvSZ11]. Our scope is much narrower: a claim-disciplined atlas centered on one proved C/F bridge.

This paper is organized around one case where the transfer mechanism is explicit. We prove and certify a direct Möbius-binomial bridge between the Zagier C and F sequences:

$$F(x) = (1 - 9x)^{-1} C\left(\frac{-x}{1 - 9x}\right).$$

The identity gives an involutive coefficient transform, an ODE pullback, and a Hankel determinant invariance, together with a constant-term complement operation. It is compatible with the known Franel network around these sequences.

The second contribution is methodological. We use the C/F bridge as a primitive in a theorem-transfer atlas for nearby Apéry-like source layers. The atlas records what can be transferred, what can only be imported by reference, what is merely metadata, and what remains a proof obligation. The point is not to claim that all adjacent congruence frameworks become theorems of this paper. Rather, we separate proof, imported theorem, finite replay, conjectural source attachment, blocked metadata, watch rows, and bounded negative evidence.

The bridge is the mathematical core; the atlas records how far its transport consequences extend. Sections 2–4 give the C/F bridge and its certificate. Sections 5–11 explain how the bridge is used to build a disciplined theorem-transfer atlas. Appendices contain the theta-gauge certificate, the Atkin–Lehner normalization audit, the constant-term complement guardrail, selected registry rows, open problems, and source-status notes.

2 Zagier C and F

Let C_n and F_n denote Zagier's C and F Apéry-like sequences in the catalog convention used throughout this paper [Zag09, OEI26a, OEI26b]. Write

$$C(x) = \sum_{n\geq 0} C_n x^n, \qquad F(x) = \sum_{n\geq 0} F_n x^n.$$

The C sequence is given by

$$C_n = \sum_{k=0}^{n} \binom{n}{k}^2 \binom{2k}{k}.$$

It satisfies

$$(n+1)^2 C_{n+1} = (10n^2 + 10n + 3)C_n - 9n^2 C_{n-1}.$$

The F sequence is normalized by $F_0 = 1$, $F_1 = 6$, and satisfies

$$(n+1)^2 F_{n+1} = (17n^2 + 17n + 6)F_n - 72n^2 F_{n-1}.$$

The atlas below always treats these catalog-normalized objects as the named source and target rows.

3 The C/F Möbius-binomial bridge

3.1 Modular parameter and theta-gauge normalization

We use the modular parametrizations recorded by Beukers in the common $\Gamma_0(6)$ setting [Beu25]. In the C row, let $t_C(q)$ denote the source modular parameter and put

$$\Theta_C(q) = C(t_C(q)).$$

For the F row, Beukers' source parameter is twist-guarded. We write it as $t_F^{\mathrm{source}}(q)$ and use the catalog parameter

$$t_F(q) = t_F^{\mathrm{cat}}(q) := -t_F^{\mathrm{source}}(q).$$

The F theta series used below is therefore

$$\Theta_F(q) = F(t_F(q)) = F(-t_F^{\mathrm{source}}(q)) = \Theta_F^{\mathrm{source}}(q).$$

All q-expansions, cusp-order checks, and Sturm checks in this paper are in this common normalization. The two F parameters are kept distinct:

$$t_F^{\mathrm{source}} = \frac{t_C}{1-9t_C}, \qquad t_F^{\mathrm{cat}} = -\frac{t_C}{1-9t_C}.$$

Consequently

$$1 + 9t_F^{\mathrm{source}} = 1 - 9t_F^{\mathrm{cat}} = \frac{1}{1-9t_C}.$$

In particular, the two identities used in the bridge proof are identities of meromorphic modular functions on $X_0(6)$ in this normalization.

Theorem 3.1 (Zagier C/F Möbius-binomial bridge)**.** *Under the catalog convention used here and the verified F twist transport,*

$$F(x) = (1-9x)^{-1} C\left(\frac{-x}{1-9x}\right).$$

Proof. With the normalization of Section 3.1, the theta-gauge identities are

$$\Theta_F/\Theta_C = 1 - 9t_C, \qquad \Theta_C/\Theta_F = 1 + 9t_F^{\mathrm{source}} = 1 - 9t_F^{\mathrm{cat}}.$$

The second form is the one compatible with the catalog parameter. With

$$u = t_C(q), \qquad x = t_F^{\mathrm{cat}}(q) = \frac{-u}{1-9u},$$

one has

$$\frac{-x}{1-9x} = u, \qquad (1-9x)^{-1} = 1 - 9u.$$

Thus

$$F(x) = \Theta_F(q) = (1-9u)\Theta_C(q) = (1-9x)^{-1} C\left(\frac{-x}{1-9x}\right).$$

The theta-gauge identities are verified by forward and inverse valence certificates, with a cleared-denominator Sturm backup; see Appendix A. The same normalization also has an Atkin–Lehner W_2 parameter and theta audit recorded in Appendix A.5; this audit is explanatory and is not used as a substitute for the Möbius-binomial bridge proof. □

Corollary 3.2 (Coefficient transform)**.** *Define*

$$T(a)_m = \sum_{n=0}^{m} \binom{m}{n} (-1)^n 9^{m-n} a_n.$$

Then $T(C) = F$ and $T(F) = C$.

Proof. Taking coefficients in Theorem 3.1 gives $T(C) = F$. Let $g(x) = -x/(1-9x)$. Then $g(g(x)) = x$ and

$$(1-9x)^{-1}(1-9g(x))^{-1} = 1.$$

Applying the same transform a second time therefore gives $T(F) = C$. □

Corollary 3.3 (Hankel determinants)**.** *For a sequence $A = (A_n)_{n\geq 0}$ put*

$$\Delta_N(A) = \det(A_{i+j})_{0\leq i,j<N}.$$

Then, for every $N \geq 1$,

$$\Delta_N(C) = \Delta_N(F).$$

Proof. This is the standard Hankel invariance of binomial transforms [SS06], but the short argument is included to fix the signs. Set $G_m = (-1)^m F_m$. By Corollary 3.2,

$$G_m = \sum_{n=0}^{m} \binom{m}{n} (-9)^{m-n} C_n.$$

For fixed N, let B_N be the lower triangular matrix

$$(B_N)_{i,j} = \begin{cases} \binom{i}{j}(-9)^{i-j}, & j \leq i, \\ 0, & j > i, \end{cases} \qquad 0 \leq i, j < N.$$

The matrix B_N has determinant 1, and Vandermonde convolution gives

$$(G_{i+j})_{0\le i,j<N} = B_N\,(C_{i+j})_{0\le i,j<N}\,B_N^t.$$

Hence $\Delta_N(G) = \Delta_N(C)$. Finally,

$$(F_{i+j})_{0\le i,j<N} = D_N\,(G_{i+j})_{0\le i,j<N}\,D_N, \qquad D_N = \operatorname{diag}((-1)^0,\dots,(-1)^{N-1}),$$

so $\Delta_N(F) = \Delta_N(G)$. □

Corollary 3.4 (ODE pullback)**.** *Let*

$$L_C = x(x-1)(9x-1)D_x^2 + (27x^2-20x+1)D_x + 3(3x-1)$$

be the differential operator associated to the C recurrence, and let

$$L_F = x(72x^2-17x+1)D_x^2 + (216x^2-34x+1)D_x + 6(12x-1)$$

be the corresponding F operator. Under

$$u = -\frac{x}{1-9x}, \qquad F(x) = (1-9x)^{-1}C(u),$$

symbolic substitution gives

$$L_C[C](u) = (9x-1)L_F[F](x).$$

Thus the C equation pulls back to the F equation up to the rational factor $9x-1$, which is nonzero as a rational function.

Remark 3.5 (Franel route)**.** *Let $R_n = \sum_{k=0}^{n}\binom{n}{k}^3$. The known Franel transforms are*

$$C(x) = (1-x)^{-1}R\left(\frac{x}{1-x}\right), \qquad F(x) = (1-8x)^{-1}R\left(\frac{-x}{1-8x}\right),$$

with provenance through Barrucand, Callan, Verrill, and OEIS [Bar75, Cal08, Ver10]. Composing these formulas gives the same C/F bridge.

4 Certificate and normalization

The proof depends on four normalization checks. First, C and F are placed in a common $\Gamma_0(6)$ normalization. Second, the source F parameter is transported to the catalog-normalized F twist. Third, the forward and inverse theta-gauge identities are certified by cusp-order and valence arguments. Fourth, a cleared-denominator Sturm check gives an independent backup. The main text uses only the resulting identities; the certificate details are recorded in Appendix A.

5 Additional functorial consequences

The C/F bridge is used functorially in a small number of controlled ways. The closed internal consequences are the coefficient involution, the Hankel determinant invariance, the ODE pullback, and the binomial-complement constant-term lemma. Together with the imported C-side Laurent model, that lemma gives the $9-\Lambda_{C,1}$ complement witness. The source-layer consequences below explain how those facts are recorded in the atlas. They do not promote adjacent Rowland–Yassawi, BTY, or Dwork comparison metadata into unconditional theorem claims.

5.1 Coefficient transform and Liu normalization

The coefficient transform in Corollary 3.2 is the atlas primitive. It allows a theorem stated for F to be rewritten as a theorem for $T(C)$. It does not automatically produce a theorem for raw C_n.

In Liu's normalization

$$v_m^{(A)}(u) = \sum_{n=0}^{m} \binom{m}{n} (-u)^{m-n} A_n$$

[Liu25], the same bridge reads $v_m^{(C)}(9) = (-1)^m F_m$ and, by involution, $v_m^{(F)}(9) = (-1)^m C_m$. This is a normalization map into Liu's binomial-transform layer, not a blanket import of every adjacent congruence.

For example, Sun's source rows include congruences for a sequence Q_n with $Q_n = (-1)^n F_n$ in the relevant convention [Sun22]. The bridge rewrites these as statements about $(-1)^n T(C)_n$. Sun's raw-C conjectural congruence is not a consequence of this C/F bridge; it is addressed separately by a Franel–Domb bridge through intermediate Franel, Domb, and Rodriguez–Villegas sums [Bab26].

5.2 Franel network

Remark 3.5 explains why the bridge is not an isolated accident: C and F are both connected to the Franel sequence by known Möbius-binomial transforms. The theorem-transfer atlas records this as known source metadata. It does not treat every inverse or coefficient replay as a new theorem edge.

The source-layer order is important: the Franel sequence is the common upstream object, while the C/F bridge is the induced relation after the two classical transforms are composed. This is why the paper can use Franel as a provenance check without making every Franel-side theorem a C/F theorem.

More concretely, both generating functions are obtained from the same cubic Franel series $R(x) = \sum R_n x^n$ by two different fractional-linear changes of variable. The C/F bridge is therefore the composition of two classical Franel transforms, not a relation inferred only from matching initial coefficients. This Franel viewpoint is useful in the atlas because it separates a known structural network from later theorem-transfer claims.

5.3 Constant-term complement

The coefficient transform has a direct constant-term interpretation.

Lemma 5.1 (Binomial-complement constant-term transform)**.** *Let Λ be a Laurent polynomial and let*

$$A_n = \mathrm{CT}(\Lambda^n).$$

For a scalar a, define

$$B_n = \sum_{k=0}^{n} \binom{n}{k} (-1)^k a^{n-k} A_k.$$

Then

$$B_n = \mathrm{CT}((a - \Lambda)^n).$$

Proof. By the binomial theorem,

$$(a - \Lambda)^n = \sum_{k=0}^{n} \binom{n}{k} a^{n-k} (-\Lambda)^k.$$

Taking constant terms gives

$$\mathrm{CT}((a - \Lambda)^n) = \sum_{k=0}^{n} \binom{n}{k} (-1)^k a^{n-k}\, \mathrm{CT}(\Lambda^k) = B_n.$$

□

For

$$\Lambda_{C,1} := \Lambda_C = \frac{(x+y+1)(xy+x+y)}{xy}$$

Gorodetsky records this Laurent model for the C row [Gor23]. Equivalently, if $P = (x+y+1)(xy+x+y)$, then

$$\mathrm{CT}(\Lambda_{C,1}^n) = [x^n y^n] P^n,$$

and the cited model identifies this coefficient with the catalog-normalized C_n. Lemma 5.1 gives

$$F_n = \mathrm{CT}((9 - \Lambda_{C,1})^n).$$

This is a constant-term operation. It is not a Laurent mutation claim and it is not a Newton-equivalence claim between $9 - \Lambda_{C,1}$ and other F-side models such as Gorodetsky's replacement model Q_F [Gor23].

5.4 Rational diagonals and small automata

The constant-term models also admit rational diagonal contracts, which support small automatic-congruence metadata for C/F and related rows. This paper records those automata as metadata. It does not claim a full Rowland–Yassawi theorem-by-reference application, because that would require the source-level bad-prime and singularity filters in addition to an explicit Cartier-state construction [RY15].

6 Claim classes and atlas discipline

The atlas uses explicit claim classes. Table 1 summarizes the ones used in this paper.

`closed_internal_result`	Proved or certified inside the C/F package.
`theorem_import_by_reference`	External theorem used with its source label; not renamed as an internal theorem.
`conditional_theorem_transfer`	Transfer valid after an explicit bridge or source assumption.
`closed_metadata`	Verified auxiliary record, such as a normalization map, fixed-modulus automaton, source-label reconciliation, or known literature identity; not used as a new theorem unless separately promoted through a proof route.
`explicit_automaton_metadata`	Explicit finite-state or prime-power automaton data recorded as metadata; in the V2 package this is treated as a closed-metadata subtype, not as a theorem-transfer class.
`finite_exact_replay`	Exact finite check used as evidence only.
`source_attached_conjecture`	Conjectural source row attached to the atlas without proof.
`proof_obligation`	A precise missing lemma, not a vague direction.
`blocked`	A known missing contract prevents theorem use.
`watch`	Interesting metadata without a promotion route.
`bounded_negative`	A bounded search failed a stated gate.

Table 1: Claim classes used in the theorem-transfer atlas.

The guardrails are simple: finite replay is evidence, not proof; coefficient matching is not theorem transfer; and external theorems remain external theorems.

7 The theorem-transfer registry

The registry is an operational table of theorem imports and bridge-generated transfers. The complete table is part of the data package; Appendix C records the schema and selected rows. The core examples are:

1. the C/F bridge, a closed internal result;
2. the coefficient involution, a closed internal result;
3. the ODE pullback, a closed internal result;
4. Sun's F-side source rows transferred only to $T(C)$;
5. Sun's raw-C target, retained in the historical V1/V2 atlas as a source-attached conjecture and proof obligation, but now marked in the paper-facing readout as addressed separately by [Bab26];
6. CT/Newton guardrails, which block mutation or Newton-equivalence claims without an explicit operation.

8 Source-layer examples

8.1 Sun layer

Sun supplies source congruences for F-side objects and related binomial sums [Sun22]. The C/F bridge transfers F-side statements to $T(C)$. It does not prove raw-C variants from C/F data. The raw-C Conjecture 3.5 is a separate Franel–Domb theorem [Bab26], and the atlas keeps that proof route separate from the C/F theorem-transfer rows.

8.2 Dwork layer

Mellit–Vlasenko give Dwork-type congruences for constant terms of powers of Laurent polynomials [MV16]. The complement model $9 - \Lambda_{C,1}$ gives an additional F-side constant-term witness. We record it as Dwork metadata by reference. We do not claim a Newton equivalence or a Laurent mutation between this complement model and the F-side model Q_F.

Thus the atlas keeps three objects separate: the C-side Laurent model, the bridge-generated complement $9 - \Lambda_{C,1}$, and independent F-side Laurent models such as Q_F. The first two give a closed C/F complement witness; comparing the complement to other F-side models remains open.

8.3 Automatic congruence layer

Rowland–Yassawi provide the source framework for automatic congruences of rational diagonals [RY15]. The atlas records explicit small automata for Cooper, Franel, and C/F small moduli, including C/F mod 2, mod 3, mod 4, and mod 9 metadata. These are explicit automaton or prime-power metadata; they are not a full Rowland–Yassawi theorem import.

8.4 Chan–Cooper companion Lucas lane

The BTY/Chan–Cooper audit separates the raw weight-one problem from a weight-two Chan–Cooper companion layer. The raw weight-one $\chi(w_d)$ values remain uncomputed. However, the five companion rows V6A, V6B, V6C, V8, and V9 admit direct Chan–Cooper binomial formulas [CC12], and their companion coefficient sequences satisfy Lucas congruences modulo every prime by a direct Lucas–Kummer argument. This result is not a C/F bridge transfer and is not a BTY Theorem 1.1 import.

Let

$$U_i(n) = \binom{2n}{n} s_i(n).$$

The source formulas are

$$s_{6A}(n) = \sum_{j=0}^{n} (-8)^{n-j} \binom{n}{j} \sum_{\ell=0}^{j} \binom{j}{\ell}^3,$$

$$s_{6B}(n) = \sum_{j=0}^{n} \binom{n}{j}^2 \binom{2j}{j}, \qquad s_{6C}(n) = \sum_{j=0}^{n} \binom{n}{j}^3,$$

$$s_8(n) = \sum_j 4^{n-2j} \binom{n}{2j} \binom{2j}{j}^2,$$

and

$$s_9(n) = \sum_j (-3)^{n-3j} \binom{n}{j} \binom{n-j}{j} \binom{n-2j}{j}.$$

The sums are taken over the natural ranges where the displayed binomial coefficients are defined. The Chan–Cooper table gives the positive-q coefficient sequence U_8; the atlas-normalized local V8 branch uses $Z_8(-q)$, hence the coefficient sequence $U_8^-(n) = (-1)^n U_8(n)$. We record both because Lucas congruences are preserved by this sign twist.

Lemma 8.1 (Lucas summation closure)**.** *Let $S(n) = \sum_j f(n,j)$ be a finite integer-valued sum, with $f(n,j) = 0$ outside its natural support. Fix a prime p and write $n = n_0 + pN$ and $j = j_0 + pJ$, with $0 \le n_0, j_0 < p$. Suppose the carry-free terms satisfy*

$$f(n_0 + pN, j_0 + pJ) \equiv f(n_0, j_0) f(N, J) \pmod p$$

and all carry cases vanish modulo p. Then

$$S(n_0 + pN) \equiv S(n_0) S(N) \pmod p.$$

Consequently, if the hypothesis holds for every prime p, then S satisfies Lucas congruences modulo every prime.

Theorem 8.2 (Chan–Cooper companion Lucas rows)**.** *The five sequences*

$$U_{6A}, U_{6B}, U_{6C}, U_8, U_9$$

satisfy Lucas congruences modulo every prime. The negative-q level-8 sequence

$$U_8^-(n) = (-1)^n U_8(n)$$

also satisfies Lucas congruences modulo every prime.

Proof. We use the standard Lucas theorem for binomial and multinomial coefficients, together with the Kummer carry criterion and Lemma 8.1.

First, the sequence a^n satisfies Lucas congruences modulo every prime. If $p \nmid a$, this follows from Fermat's theorem; if $p \mid a$, the sequence is congruent to $1, 0, 0, \dots$ modulo p. Products of Lucas sequences are Lucas, and the central binomial sequence $\binom{2n}{n}$ is Lucas, with digit-doubling carries producing zero factors.

The Franel cubic sum

$$R(n) = \sum_j \binom{n}{j}^3$$

is Lucas by applying Lucas theorem termwise. This proves the claim for s_{6C}. The same argument, with the central binomial factor $\binom{2j}{j}$, proves the claim for

$$s_{6B}(n) = \sum_j \binom{n}{j}^2 \binom{2j}{j}.$$

For s_{6A}, write

$$s_{6A}(n) = \sum_j \binom{n}{j} R(j)(-8)^{n-j}.$$

This is a binomial convolution of Lucas sequences, hence is Lucas.

For s_8, when p is odd, Lucas and Kummer factor the summand

$$4^{n-2j}\binom{n}{2j}\binom{2j}{j}^2$$

digitwise; carry cases vanish modulo p. For $p = 2$, one has $s_8(0) = 1$ and $s_8(n) \equiv 0 \pmod 2$ for $n > 0$, since the $j = 0$ term contains 4^n and $\binom{2j}{j}$ is even for $j > 0$.

For s_9, when $p \neq 3$, the multinomial form

$$(-3)^{n-3j}\frac{n!}{j!j!j!(n-3j)!}$$

factors digitwise in carry-free cases, and carry cases vanish modulo p. For $p = 3$, if $n > 3j$, the factor $(-3)^{n-3j}$ is divisible by 3. If $n = 3j$ and $j > 0$, then $(3j)!/(j!^3)$ is divisible by 3, since adding $j + j + j$ in base 3 creates a carry. Thus $s_9(0) = 1$ and $s_9(n) \equiv 0 \pmod 3$ for $n > 0$.

Multiplication by $\binom{2n}{n}$ preserves Lucas congruences, so the five $U_i(n)$ are Lucas. Finally, $(-1)^n$ is Lucas for odd primes and is invisible modulo 2, so $U_8^-(n) = (-1)^n U_8(n)$ is Lucas as well. $\square$

This theorem is independent of the C/F bridge. It is also independent of the BTY theorem-transfer mechanism. The level-6 rows have a separate local BTY/Atkin–Lehner audit, and the V8 and V9 normalizer geometry remains useful modular context, but neither is used in the proof above. No raw weight-one $\chi(w_d)$ values are computed here.

Other source layers, including Liu's binomial-transform congruence profile, Straub's Gessel–Lucas theorem, and Beukers–Tsai–Ye's modular-form Lucas congruence source [Liu25, Str24, BTY25], are summarized in Appendix E. They are kept brief here because their role is source attachment and claim discipline, not the main C/F theorem.

9 Open-problem atlas

The open-problem atlas records blockers rather than vague directions. A selection is shown in Table 2; a fuller schema appears in Appendix D.

10 Edge scan and watch-triage discipline

A bounded C/F-like Möbius-binomial scan checked 17684 transforms of the form

$$B(x) = (1-ax)^{-r}A\left(\frac{sx}{1-ax}\right).$$

The scan used exact integer arithmetic on 12 coefficients, the 16 catalog families recorded in the companion data, $r \in \{1, 2, 3\}$, $-200 \le a \le 200$, and nonzero integers s with $|s| \le 20$. It recovered known or canonical relations and promoted no new theorem edge. The eight previous watch hits were subsequently resolved as known metadata: six are alias/canonical identity hits for the Apéry zeta(3) and Almkvist–Zudilin gamma catalog rows, and two are inverse Franel-transform hits. This is a negative discipline result: coefficient matches are not automatically promoted.

Branch	Blocking lemma or missing contract	Status
BTY weight-one rows	row-specific $\chi(w_d)$ values	open
Straub derivative transfer	low-carry identity for the corrected connection	proof obligation
Dwork model comparison	controlled equivalence between complement model and Q_F	open
Rowland–Yassawi source application	bad-prime/singularity filter and source-style Cartier states	inactive
Edge-watch scan	no unresolved V1 watch rows remain	closed metadata
Sun raw-C to RV-108	Franel–Domb raw-C/RV bridge outside the C/F atlas	companion manuscript
A2b global glue	Domb branch-multiplier identity outside the C/F atlas	companion manuscript

Table 2: Selected open-problem and blocker rows.

11 Data package and reproducibility

The data package contains registry rows, claim classes, source labels, proof obligations, open-problem statuses, edge-scan triage records, manifests, and readouts. The paper-facing purpose is reproducibility of claim status: a reader should be able to tell which rows are proved internally, which are imported by reference, which are finite replays, and which are blocked. The package is based on the V2 manifest, with the V1 registry retained as immutable provenance for the original C/F theorem-transfer rows. Appendix E lists source identifiers and package fields. The curated companion package is available at

`https://github.com/aconsciousfractal/C-F_Bridge_and_Atlas`,

with the paper source in `paper/`, JSON artifacts in `data/relation_atlas/`, replay scripts in `scripts/`, and the executable-scope note in `docs/atlas/EXECUTABLE_REPLAY_SCOPE.md`.

12 Conclusion

We do not solve every problem attached to the C/F bridge. Instead, we give a normalized proof of the bridge and a reproducible model for using it without overclaiming. The bridge transports generating functions, coefficients, ODEs, Hankel determinants, and a constant-term complement model. The atlas records adjacent theorem sources, metadata layers, finite checks, conjectures, blockers, and bounded negatives with distinct claim classes. This separation is the main lesson: bridge identities are powerful precisely when their scope is explicit.

A Theta-gauge certificate details

This appendix records the certificate data behind the theta-gauge identities used in Theorem 3.1.

By the normalization contracts in Section 3.1, the functions below are meromorphic modular functions on $X_0(6)$ and have no interior poles. Their only possible poles are cuspidal; the cusp-order tables bound those poles. The certificate uses the following public data contract.

object	expression	weight	character	audit source
t_C	Beukers $X_0(6)$ parameter	0	1	cusp orders and q-replay
t_F^{source}	$t_C/(1-9t_C)$	0	1	twist-source contract
t_F^{cat}	$-t_F^{\text{source}}$	0	1	catalog twist contract
Θ_F/Θ_C	$1-9t_C$	0	1	forward valence certificate
Θ_C/Θ_F	$1-9t_F^{\text{cat}}$	0	1	inverse valence certificate

The checked q-expansion orders are recorded here as certificate data. The public companion scripts reproduce the finite coefficient and atlas checks; larger q-expansion/Sturm certificate automation is outside the executable scope declared in the companion repository.

A.1 Forward valence certificate

The forward identity is

$$\Theta_F/\Theta_C = 1 - 9t_C.$$

Define

$$H(q) = \Theta_F(q)/\Theta_C(q) - (1 - 9t_C(q)).$$

The cusp-order audit on $\Gamma_0(6)$ gives

object	0	1/2	1/3	∞
t_C	0	-1	0	1
Θ_F/Θ_C	1	-1	0	0.

Thus the only possible pole of H is at the cusp 1/2, of order at most 1. The q-expansion replay gives $\mathrm{ord}_\infty(H) \geq 201$. Since this exceeds the possible pole order, the valence criterion forces $H = 0$.

A.2 Inverse valence certificate

The inverse identity, written in the catalog parameter, is

$$\Theta_C/\Theta_F = 1 - 9t_F^{\mathrm{cat}}.$$

Define

$$K(q) = \Theta_C(q)/\Theta_F(q) - (1 - 9t_F^{\mathrm{cat}}(q)).$$

The inverse cusp-order audit gives

object	0	1/2	1/3	∞
t_F^{cat}	-1	0	0	1
Θ_C/Θ_F	-1	1	0	0.

Thus the only possible pole of K is at the cusp 0, of order at most 1. The q-expansion replay gives $\mathrm{ord}_\infty(K) \geq 201$, and the valence criterion gives $K = 0$.

A.3 Sturm clear-denominator backup

The same two identities are also checked after clearing the possible cuspidal poles. Use the eta quotient

$$M(q) = \frac{\eta(q)^3\eta(2q)^3}{\eta(3q)\eta(6q)}.$$

By the Ligozat criterion, the exponent vector of M satisfies the two congruences modulo 24 and the total weight is 2. The cusp orders of M on $\Gamma_0(6)$ are

cusp	0	1/2	1/3	∞
$\mathrm{ord}(M)$	1	1	0	0.

Thus M is holomorphic at every cusp and vanishes at the two cusps where a pole can occur in the preceding valence argument. The deduction is explicit:

- H has no possible pole except at 1/2, of order at most 1, while $\mathrm{ord}_{1/2}(M) = 1$; hence MH is holomorphic at 1/2.

- At the other cusps H is holomorphic and M is holomorphic, so MH is holomorphic there as well.
- K has no possible pole except at 0, of order at most 1, while $\mathrm{ord}_0(M) = 1$; hence MK is holomorphic at 0.
- At the other cusps K is holomorphic and M is holomorphic, so MK is holomorphic there as well.

Since H and K have weight 0 and trivial character in the chosen normalization, both MH and MK are holomorphic weight-2 forms on $\Gamma_0(6)$ with the character of the eta quotient M. The Sturm criterion used here is the standard one for the corresponding character space; their checked q-expansions lie in the rational coefficient field used by the certificate, so the ordinary coefficient-vanishing test applies to the cleared forms. Since

$$[\mathrm{SL}_2(\mathbb{Z}) : \Gamma_0(6)] = 12, \qquad \left\lfloor \frac{2 \cdot 12}{12} \right\rfloor = 2,$$

the Sturm bound is 2. The replay gives vanishing at infinity far beyond this bound, hence both cleared forms vanish identically.

A.4 F twist transport dependency

The F-side parameter in source notation is twist-guarded. If $F(x) = \sum_{n\geq 0} F_n x^n$ denotes the catalog-normalized generating function, the transport contract is

$$F\big(-t_F^{\mathrm{source}}(q)\big) = \Theta_F^{\mathrm{source}}(q),$$

or equivalently

$$t_F^{\mathrm{cat}}(q) = -t_F^{\mathrm{source}}(q).$$

This is why Theorem 3.1 is stated in catalog-normalized F notation and why the inverse theta-gauge identity is written as $\Theta_C/\Theta_F = 1 - 9t_F^{\mathrm{cat}}$.

A.5 Atkin–Lehner W2 parameter and theta audit

The C/F normalization also admits a precise Atkin–Lehner description. Put

$$h = t_F^{\mathrm{source}}, \qquad t_C = \frac{h}{1+9h}, \qquad t_F^{\mathrm{cat}} = -h,$$

and let

$$W_2 = \begin{pmatrix} 2 & -1 \\ 6 & -2 \end{pmatrix}.$$

For weight-zero functions, $f|W$ means precomposition with W. For a weight-one form we use the convention

$$(f|_1W)(\tau) = \det(W)^{1/2}(c\tau + d)^{-1} f(W\tau),$$

with the positive real branch $\det(W_2)^{1/2} = \sqrt{2}$.

Lemma A.1 (Atkin–Lehner W_2 audit)**.** *In the above normalization,*

$$t_C|W_2 = 1 + 8t_F^{\mathrm{source}} = 1 - 8t_F^{\mathrm{cat}},$$

and the theta eta quotients satisfy

$$\Theta_C|_1W_2 = (2\sqrt{2})^{-1}\Theta_F, \qquad \Theta_F|_1W_2 = -2\sqrt{2}\,\Theta_C.$$

Proof. The parameter identity follows by substituting the normalized action

$$W_2(h) = -\frac{h+1/8}{9(h+1/9)}$$

into $t_C = h/(1+9h)$. This gives $t_C|W_2 = 1+8h$.

For the theta statement, write $\eta_d(\tau) = \eta(d\tau)$. The eta quotients are

$$\Theta_C = \frac{\eta_2^6\eta_3}{\eta_1^3\eta_6^2}, \qquad \Theta_F = \frac{\eta_1^6\eta_6}{\eta_2^3\eta_3^2}.$$

The four eta arguments $dW_2\tau$, for $d \in \{1,2,3,6\}$, factor projectively through $\mathrm{SL}_2(\mathbb{Z})$ matrices as follows:

d	γ_d	factorization	target eta index
1	$\begin{pmatrix} 1 & -1 \\ 3 & -2 \end{pmatrix}$	$\gamma_1\ \mathrm{diag}(2,1)$	2
2	$\begin{pmatrix} 2 & -1 \\ 3 & -1 \end{pmatrix}$	$\gamma_2\ \mathrm{diag}(1,1)$	1
3	$\begin{pmatrix} 1 & -3 \\ 1 & -2 \end{pmatrix}$	$\gamma_3\ \mathrm{diag}(6,1)$	6
6	$\begin{pmatrix} 2 & -3 \\ 1 & -1 \end{pmatrix}$	$\gamma_6\ \mathrm{diag}(3,1)$	3.

Using the standard Dedekind eta multiplier formula

$$\eta(\gamma\tau) = \exp(\pi i R(\gamma))(c\tau+d)^{1/2}\eta(\tau), \qquad R(\gamma) = \frac{a+d}{12c} - s(d,c) - \frac{1}{4},$$

the exact exponents are

d	1	2	3	6
$R(\gamma_d)$	$-1/3$	$-1/6$	$-1/3$	$-1/6$.

For Θ_C, with eta exponent vector $(-3,6,1,-2)$, the total root-of-unity exponent is

$$-3(-1/3) + 6(-1/6) + 1(-1/3) - 2(-1/6) = 0.$$

For Θ_F, with exponent vector $(6,-3,-2,1)$, it is

$$6(-1/3) - 3(-1/6) - 2(-1/3) + 1(-1/6) = -1.$$

The remaining τ-dependent factors cancel against the weight-one slash prefactor. The residual powers of 2 are $2^{-3/2}$ for Θ_C and $-2^{3/2}$ for Θ_F, giving the two displayed identities. □

This audit is a normalization statement, not a replacement for the bridge. Indeed, in the coordinate $T = t_C$,

$$T|W_2 = \frac{1-T}{1-9T}, \qquad g(T) = -\frac{T}{1-9T},$$

so the catalog C/F bridge map is not literally the Atkin–Lehner action. Rather,

$$g(T) = \frac{1-(T|W_2)}{8}.$$

Thus the F catalog parameter is an affine normalization of the W_2-translate of the C parameter. This observation also does not compute the row-specific BTY multipliers $\chi(w_d)$, which remain outside the closed claims of this paper.

B Binomial-complement constant-term lemma

Lemma 5.1 is included in the main text because it is the cleanest bridge between coefficient transforms and constant-term models. The important guardrail is repeated here: the operation

$$\Lambda \mapsto a - \Lambda$$

preserves the fact that one has a constant-term sequence, but it is not a Laurent mutation claim and does not imply Newton equivalence between two chosen Laurent models.

C Registry schema and selected rows

Each registry row uses the following schema:

`statement_id`	row identifier
`source_paper`	external or internal source
`source_label`	the theorem, conjecture, or artifact label
`source_object`	source sequence or model
`target_object`	target sequence or model
`rewrite`	transport or normalization formula
`cf_role`	how the C/F bridge enters
`claim_level`	claim class
`finite_replay`	finite evidence, if any
`literature_status`	source status
`proof_obligation`	missing lemma, if any
`status`	current row status
`next_action`	next audit/proof step.

A representative public excerpt is:

`statement_id`	source	target	rewrite/status	claim level
`cf_bridge`	C	F	$F = \mathcal{T}(C)$	closed internal result
`cf_involution`	C,F	F,C	$\mathcal{T}^2 = 1$	closed internal result
`cf_hankel`	C,F	C,F	$\Delta_N(C) = \Delta_N(F)$	closed internal result
`cf_ode_pullback`	L_C	L_F	rational pullback	closed internal result
`sun_f_to_tc`	Sun F-side	$T(C)$	signed transfer only	conditional transfer
`sun_raw_c`	raw C	RV-108	no C/F transfer	separate proof obligation
`dwork_ct`	$\Lambda_{C,1}$	$9 - \Lambda_{C,1}$	CT complement	closed metadata
`ry_automata`	small moduli	automata	fixed-modulus data	closed metadata
`chan_cooper` `five_companion` `lucas`	Chan–Cooper	U_i rows	direct Lucas proof	closed internal result

The frozen V1 machine-readable table is the JSON file

`data/relation_atlas/`
`cf_theorem_transfer_registry_v1.json`.

The June 2026 paper-facing addendum rows, including Hankel, Atkin–Lehner W_2, and Chan–Cooper Lucas, are recorded separately in

`data/relation_atlas/`
`cf_theorem_transfer_registry_`
`2026_06_extension.json`

and summarized in the V2 registry/readout documents. The excerpt is included only to make the atlas semantics visible in the paper.

D Open problems and blocking lemmas

The atlas records open work by the mathematical object that is missing. For the Sun branch, the C/F theorem transports F-side statements to $T(C)$, but it does not itself supply the raw-C/RV-108 bridge. In the historical V1/V2 data package this appears as a deferred proof obligation. After that atlas freeze, the missing raw-C/RV bridge and its A2b Domb component are recorded in the separate Franel–Domb bridge manuscript on Sun's Conjecture 3.5 [Bab26]. We keep the registry row as provenance for the C/F atlas rather than reclassifying it as a C/F consequence.

The modular-form and constant-term branches have different blockers. The Beukers–Tsai–Ye weight-one rows require row-specific multiplier data $\chi(w_d)$ before they can be used as theorem imports. The Dwork branch has a proved C/F constant-term complement operation, but the stronger comparison between the complement model and the Gorodetsky F-side model Q_F remains an equivalence-or-obstruction problem. The Rowland–Yassawi branch has explicit small automata as metadata, but a full source-theorem application would still require a bad-prime/singularity filter and source-style Cartier states.

Finally, the Straub derivative-transfer branch is not closed by the formal C/F coefficient involution alone. The corrected connection term leaves a finite low-carry identity as the remaining proof obligation. The edge-watch scan no longer contributes an open problem: its eight watch rows have been resolved as known metadata.

E Source and bibliography status

Zagier and OEIS fix the catalog-normalized C and F rows used throughout the paper. Beukers supplies the modular parametrizations, the common $\Gamma_0(6)$ setting, and the F twist caveat used in the theta-gauge certificate. Barrucand, Callan, and Verrill supply the classical Franel transform context that explains the C/F bridge as part of a known network rather than as an isolated coefficient coincidence.

The remaining sources enter as theorem imports or metadata layers. Sun supplies the source congruence layer and the original Conjecture 3.5 target [Sun22]; the latter is addressed in a separate Franel–Domb bridge manuscript rather than by the C/F bridge [Bab26]. Liu supplies binomial-transform congruence metadata with convention guards [Liu25]. In the Liu audit, thirteen rows match the normalized catalog directly, while the η and $s18$ rows are stored with an explicit convention map rather than as failures. Chan–Cooper supplies the companion binomial formulas for the five rows in Theorem 8.2; the Lucas proof there is internal and does not import a BTY theorem [CC12]. Straub's Gessel–Lucas theorem is imported by reference where its hypotheses are used, but the separate C/F derivative-transfer problem remains guarded by its low-carry proof obligation [Str24].

The Beukers–Tsai–Ye theorem is used only when row hypotheses are complete [BTY25]. Mellit–Vlasenko and Gorodetsky provide the Dwork and constant-term context, and Rowland–Yassawi provides the source contract for automatic congruences [MV16, Gor23, RY15]. The broader tabular background of Calabi–Yau and Apéry-like operators is represented by the Almkvist–van Enckevort–van Straten–Zudilin tables and the subsequent transformation literature [AvEvSZ05, AvSZ11]. Before submission, the remaining bibliographic work is journal-specific style and final access-date policy for web catalog entries.

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